Akshay Meshiya

Process Engineer
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PROFESSIONAL SUMMARY

Process engineer with an M.S. in Chemical Engineering from Rowan University and 6+ years in process engineering, batch and continuous chemical production, and industrial wastewater treatment in the US and India. My work covers process design and development, process optimization, scale-up and technology transfer, PFD/P&ID development, material and energy balances, and troubleshooting. I use SPC, root cause analysis (RCA), and CAPA to fix problems at the source, and I work within HAZOP, process safety management (PSM), and management of change (MOC). Results so far include fewer batch failures, lower chemical use, and faster reporting, achieved through cross-functional work with operations, maintenance, quality, and EHS.

WORK EXPERIENCE

Process Engineer **Apr 2026 – Present**
Valcor Engineering Corporation **NJ, USA**

- Model high-purity chemical processes in Aspen Plus, Aspen HYSYS, and ChemCAD, improving solvent recovery efficiency by 7% while meeting environmental compliance standards.
- Develop PFDs, P&IDs, and 3D plant layouts in AutoCAD Plant 3D and SmartPlant P&ID for 12 equipment upgrades that raised unit production capacity by 10%, managed through management of change (MOC).
- Reduce process variability in batch and continuous operations with statistical process control (SPC) and downtime analysis, raising throughput while holding cleanroom-grade purity.
- Build and maintain OSIsoft PI System and Power BI dashboards that track critical KPIs and process trends in real time, speeding up fault detection and cutting unplanned downtime by 15%.
- Support $2.5M+ in CAPEX projects by commissioning ultra-high-purity (UHP) gas and liquid delivery systems and linking them to SAP PP/PM for maintenance visibility and equipment traceability.
- Lead HAZOP and LOPA studies with root cause investigations, identifying 10+ safety and operational risks and keeping the site audit-ready under strict regulatory requirements.

Assistant Production Engineer **Feb 2022 – Jun 2024**
R.R. Chemical (ISO 9001:2015 Certified) **Gujarat, India**

- Developed and improved batch chemical manufacturing processes through hands-on process analysis.
- Moved new products from R&D and pilot scale into full-scale production as part of scale-up and technology transfer.
- Dug into batch interruptions and deviations with root cause analysis (RCA, 5 Whys) and put corrective and preventive actions (CAPA) in place, reducing batch failures by 15%.
- Fixed process and equipment problems on reactors, pumps, and process lines to keep production running.
- Ran quality control (QC) checks on purity, yield, and specifications under ISO 9001.
- Kept MSDS/SDS, SOPs, and EHS documents current with the safety and compliance team.

Process Executive & Chemist **Jul 2018 – Feb 2022**
Palsana Enviro Protection Ltd. (PEPL) — Common Effluent Treatment Plant **Gujarat, India**

- Ran shift operations at a common effluent treatment plant (CETP) handling wastewater from textile units, across primary, secondary, and tertiary treatment.
- Tested inlet, in-process, and treated effluent for pH, COD, BOD, TSS, TDS, color, oil & grease, MLSS, and SVI.
- Used jar tests to fine-tune coagulant, flocculant, lime, and polyelectrolyte doses, cutting chemical consumption by 8% and lowering treatment cost per unit of effluent.
- Tracked the activated sludge process (MLSS, DO, F/M ratio, settleability) and recommended adjustments when biological treatment drifted.
- Checked treated effluent against GPCB/CPCB discharge norms and helped prepare compliance reports and records for regulatory inspections.
- Replaced handwritten logbooks with Microsoft Excel trackers and trend charts for lab results, chemical use, and flow, which cut shift reporting and MIS preparation time by 30%.
- Raised plant capacity utilization by 10% by keeping treatment units within design limits, so the CETP could take effluent from more member industries.
- Watched data trends to catch process deviations early, which helped operations avoid downtime and off-spec discharge.

SKILLS

- **Process Engineering & Design:** Process Design, Process Development, Process Optimization, Scale-Up, Technology Transfer, PFD/P&ID Development, Material & Energy Balances, Heat & Mass Transfer, Equipment Sizing, Distillation, Reaction Engineering, Process Control (PID Loops, DCS/PLC Fundamentals)
- **Process Simulation & CAD:** Aspen Plus, Aspen HYSYS, ChemCAD, AutoCAD, AutoCAD Plant 3D, SmartPlant P&ID
- **Operations & Quality:** Batch & Continuous Operations, Commissioning, Process Troubleshooting, Statistical Process Control (SPC), CAPA, ISO 9001, Quality Control, KPI Monitoring, Yield Improvement, Cost Reduction
- **Process Safety & Compliance:** HAZOP, LOPA, OSHA Process Safety Management (PSM), Risk Assessment, Root Cause Analysis (RCA), Management of Change (MOC), MSDS/SDS, SOPs, EHS, Regulatory Compliance
- **Environmental:** Wastewater Treatment, Effluent Analysis, Environmental Monitoring, Pollution Control
- **Data & Tools:** Data Analysis, Trend Analysis, Python (Pandas, Matplotlib), Minitab, SAP (PP/PM), OSIsoft PI System, Power BI (PP/PM), OSIsoft PI System, Power BI, Microsoft Excel (Advanced), Microsoft Office

PROJECTS

Manufacturing of Ethylamine — B.E. Final Year Design Project **2021**

- Designed a conceptual 100 TPD ethylamine production facility; performed material and energy balances, equipment sizing, reactor and distillation design, cost estimation, and process safety evaluation.

Manufacturing of Ethylene Glycol — Diploma Final Year Design Project **2018**

- Executed process design, feasibility analysis, material balance calculations, utility assessment, and process economics evaluation.

EDUCATION

Master of Science, Chemical Engineering — Rowan University, Glassboro, NJ **Aug 2024 – May 2026**

Relevant Coursework: Process Optimization, Advanced Reactor Design, Transport Phenomena, Advanced Engineering Thermodynamics, Polymer Engineering, Water & Environment Monitoring, Sustainability Assessment, Strategic Engineering Management

Bachelor of Engineering, Chemical Engineering — Gujarat Technological University **Jul 2018 – Jul 2021**

Relevant Coursework: Process Equipment Design I & II, Mass Transfer Operations I & II, Chemical Reaction Engineering I & II, Process Heat Transfer, Fluid Flow Operations, Chemical Engineering Thermodynamics I & II, Instrumentation & Process Control, Process Modeling, Simulation & Optimization, Multicomponent Distillation, Plant Design & Project Engineering, Pollution Control, Safety & Health Management

Diploma, Chemical Engineering — Gujarat Technological University **Jul 2015 – Jun 2018**

Relevant Coursework: Mechanical Operations, Process Heat Transfer, Mass Transfer, Industrial Stoichiometry, Chemical Process Technology I & II, Utilities & Instrumentation in Chemical Plant, Pollution Control & Effluent Treatment, Safety & Hazard Management in Chemical Industry, Petroleum Refining & Petrochemical Technology